

COLLEGE PROJECT DEMONSTRATION • TEAM B

FINCORE

SECURE DIGITAL BANKING MANAGEMENT SYSTEM

Secure and Integrated Digital Banking Platform

ONE PLATFORM / CONNECTED SERVICES / SECURE FLOWS

TEAM MEMBERS



Bharathi Bhat



Chava Ramya



Shanmukha Sai



Jahnvi Pandey



Indu Patil



Raziya

TECHNOLOGIES USED

FRONTEND / ANGULAR

Bharathi • Ramya • Shanmukha • Raziya

UI development; dashboards & forms;
account, transaction, loan and EMI screens;
REST API integration.

BACKEND / SPRING BOOT

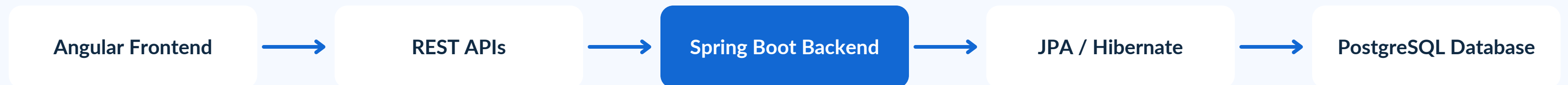
Jahnvi Pandey

REST APIs; business logic; authentication &
authorization; module integration; frontend-
database communication.

DATABASE / POSTGRESQL

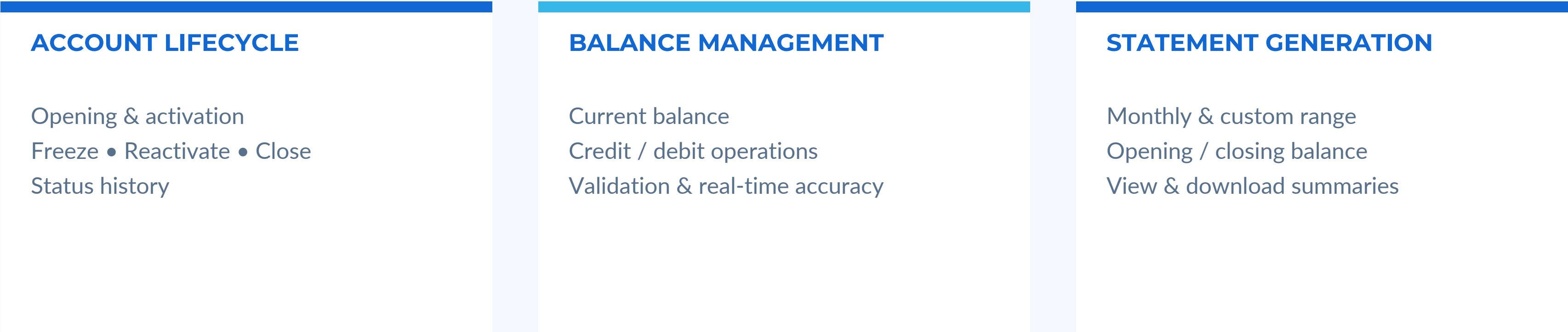
Indu Patil

Database design; tables & relationships; SQL
queries; integrity and transaction
management.



All modules are integrated into a single digital banking platform.

MILESTONE 1 – CORE BANKING COMPONENTS



MILESTONE 2 – LOAN MANAGEMENT COMPONENTS

EMI CALCULATION

$$EMI = P \times R \times (1 + R)^N / ((1 + R)^N - 1)$$

P = Principal amount • R = Monthly interest rate
N = Tenure in months

LOAN DISBURSEMENT
Approved loan processing, transfer, tracking, records, balance update and notifications.

COLLECTIONS
EMI tracking, payments, outstanding updates, partial payment, overdue handling and reminders.

OUTPUTS / Monthly EMI • Total interest • Total repayment • Schedule • Paid / Pending / Overdue



MILESTONE 3 – SECURITY & TRANSACTION SERVICES

FRAUD DETECTION

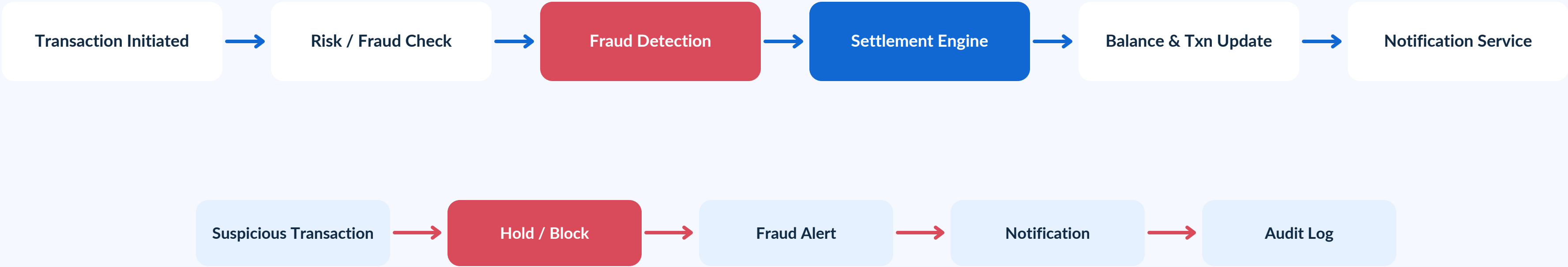
Monitor patterns, generate alerts, review and hold/block suspicious transactions; maintain fraud records.

SETTLEMENT ENGINE

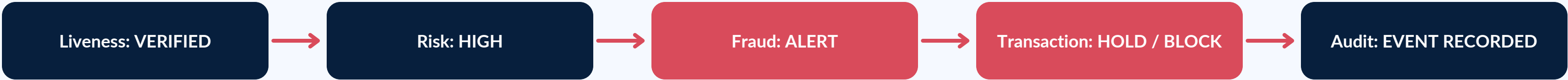
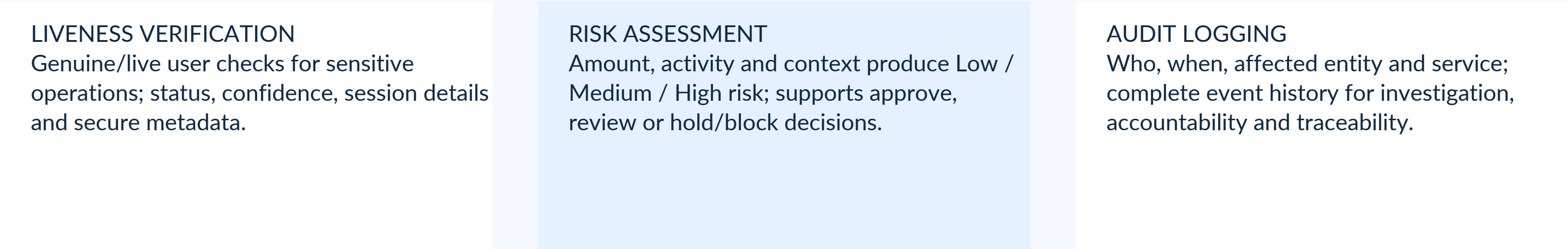
Validate and process approved transactions; record success, failure and reversals; synchronize status.

NOTIFICATION SERVICE

Success/failure, security alerts, loan and EMI reminders via Email, SMS and Push; history and status.



MILESTONE 4 – SECURITY & MONITORING COMPONENTS



OVERALL SYSTEM ARCHITECTURE

ANGULAR FRONTEND

Login • Dashboard • Accounts • Transactions • Loans / EMI • Statements • Notifications • Security Verification



SPRING BOOT BACKEND / CENTRAL INTEGRATION & BUSINESS LAYER

Authentication & Roles • Account Lifecycle • Balance • Transactions • Statements • Loans • EMI • Disbursement • Collections • Risk • Fraud • Settlement • Liveness • Notifications • Audit

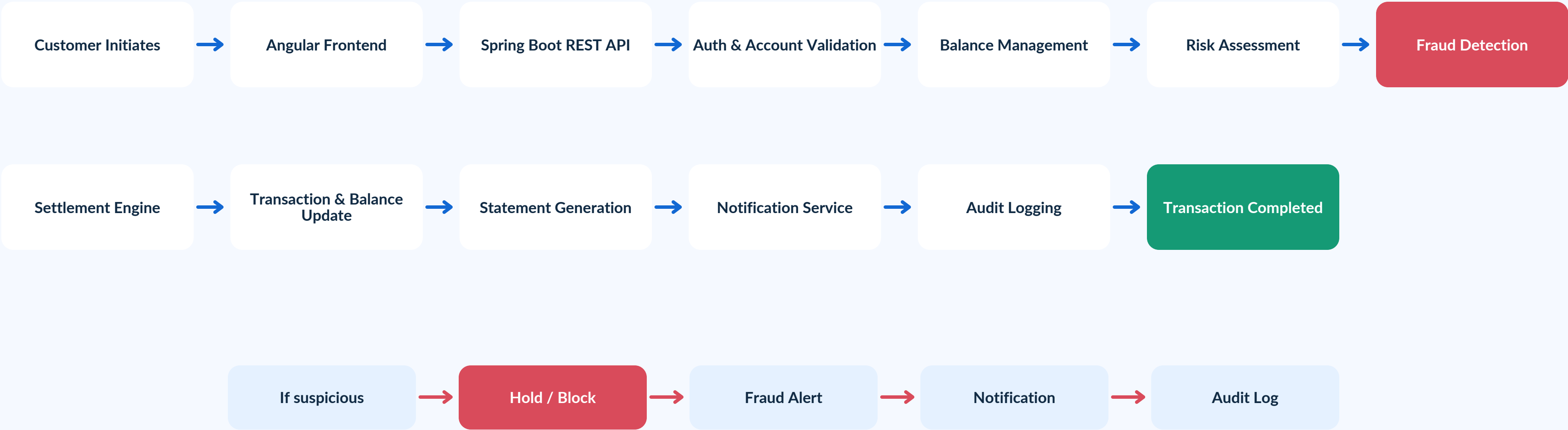


POSTGRESQL DATABASE

Users / Customers • Accounts • Transactions • Loans / EMI • Disbursements / Collections • Risk / Fraud • Settlements • Notifications • Liveness • Audit Logs

Spring Boot is the central integration layer connecting Angular with PostgreSQL and coordinating all banking services.

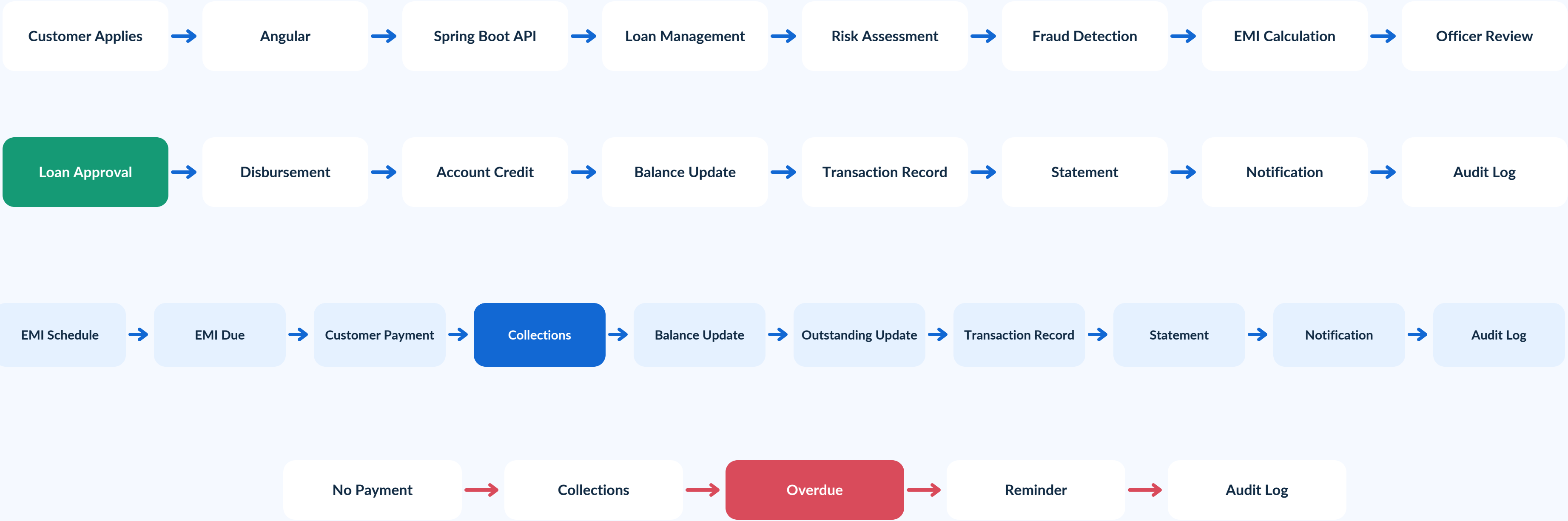
END-TO-END TRANSACTION INTEGRATION WORKFLOW



Spring Boot automatically orchestrates the required services; the customer does not manually access internal modules.

ANGULAR → REST API → SPRING BOOT → JPA / HIBERNATE → POSTGRESQL

END-TO-END LOAN INTEGRATION WORKFLOW



ANGULAR → SPRING BOOT → POSTGRESQL

CONCLUSION

FINCORE – SECURE DIGITAL BANKING MANAGEMENT SYSTEM



FINCORE demonstrates secure, connected, and end-to-end digital banking management.

THANK YOU

TEAM B

Questions & Discussion

FINCORE / SECURE DIGITAL BANKING MANAGEMENT SYSTEM